

Luca Bonaldo

Software Engineer

Contact me
me@lucabonaldo.dev

EXPERIENCE

Waterjade, Pergine (TN) — *Lead Software Engineer*

MAY 2024 - PRESENT

Leading development of **scalable software solutions**, **modernized architectures**, optimized **IoT systems**, and driving **big data** processing, aligning technical innovations with business goals.

WorldMatch, Padua (PD) — *Software Engineer*

SEPTEMBER 2022 - MAY 2024

Designed and optimized **back-end systems** for online gaming platforms, ensuring **high scalability**, **reliability**, and **seamless integration** with front-end and third-party services.

IIS Veronese-Marconi, Cavarzere (VE) — *Teacher*

DECEMBER 2023 - MARCH 2024

Focused on providing practical knowledge and fostering a **supportive environment**, helping students develop both **technical skills** and a **strong sense of collaboration**.

EDUCATION

University of Padua, Padua — *Computer Engineer (108/110)*

OCTOBER 2022 - JULY 2025

Completed a degree in **Computer Engineering**, gaining a broad understanding of **computer systems**, **software development**, **data analysis**, **networking**, and **hardware integration**, while developing strong skills in tackling **complex engineering problems**.

Liceo Scientifico G. Veronese, Chioggia — *100 with honors*

SEPTEMBER 2017 - JULY 2022

Completed a rigorous scientific curriculum with a focus on mathematics, physics, and computer science, graduating **with honors**.

SKILLS

Software **architecture** design

Cloud computing (AWS, Azure, Google Cloud)

API development

DevOps (CI/CD pipelines, Docker)

Back-end development (Python, C#, Java, Node.js)

Full-stack development

IoT integration

Database management (SQL, NoSQL)

Big data processing

AI/ML fundamentals

Problem-solving

Adaptability

Teamwork

Project management

Leadership

Strategic planning

LANGUAGES

Italian - Native

English - C2

German - A2

Spanish - A2

PROJECTS

Daqard — *Fidelity with a cue*

JULY 2025 - OCTOBER 2025

Collaborated on the design and implementation of a **B2B SaaS** platform for loyalty and fidelity program management. Contributed to the definition of a **scalable, cloud-ready architecture**, ensuring **seamless API integration** and smooth onboarding for diverse **enterprise clients**. Supported backend development initiatives focused on **performance optimization**, **system reliability**, and **long-term maintainability**, driving alignment between technical excellence and business scalability.

Nautilus — *Cost-Efficient Active Bathymetric Mapping Using Autonomous Surface Vessels*

MARCH 2025 - JULY 2025

Engineered an **adaptive algorithm** leveraging **information theory**, **Gaussian Processes**, and **machine learning** to optimize autonomous surface vessel trajectories for **efficient** underwater mapping. Developed **real-time** data analysis and path-planning modules, enhancing **survey precision** and operational **cost efficiency**. Integrated **advanced mathematical modeling**, **data-driven** decision making, and **autonomous** navigation strategies to push the boundaries of intelligent marine exploration.

INTERESTS

Machine Learning & AI

Passionate about pushing the boundaries of what **intelligent systems** can do. I actively explore advanced **machine learning** techniques, **adaptive algorithms**, and **AI-driven decision-making**, constantly experimenting to solve **complex, real-world** problems **efficiently** and **creatively**.

Cloud & Scalable Software Systems

Deeply engaged in designing and building **high-performance, cloud-native** architectures. I thrive on creating scalable, **modular** systems that are **robust, maintainable**, and capable of handling demanding workloads, always seeking **innovative solutions** to optimize performance and **reliability**.

Mathematics & Problem Solving

Enthusiastic about applying **mathematical reasoning** and **algorithmic thinking** to solve **challenging, practical** problems. I enjoy tackling challenging problems, optimizing solutions, and leveraging quantitative methods to drive **innovation** and **efficiency** in software and systems design.

METHODOLOGIES & PRACTICES

Agile, Scrum, Kanban

Test-Driven Development (TDD) & Behavior-Driven Development (BDD)

Event-Driven & Serverless Architecture

API-First & Microservices Design

Data Pipeline Design & ETL Optimization

Scalable System Design & Performance Tuning

ADDITIONAL COMPETENCIES

Technical Mentoring & Knowledge Sharing

Cross-Functional Team Collaboration

Strategic Technology Planning

Data-Driven Decision Making

Rapid Prototyping & MVP Development

Research & Proof-of-Concept Implementation